



BOSTON COLLEGE

Department of Economics

140 Commonwealth Avenue
Chestnut Hill, Massachusetts 02467

ECON 3389.3

Machine Learning for Economics

Fall 2026

Course Information

Course Title	Machine Learning for Economics
Course Number	ECON 3389
Instructor	Paul D. McNelis, S.J. ('70-MCAS)
Email	paul.mcnelis@bc.edu
Office	O'Neill 563 (south end; enter door 528, turn left at the first corridor, office at the end on the right)
Office Hours	Monday and Wednesday, 3:30–5:30 PM
Graduate Assistant	David Belyn, belyn@bc.edu
Class Time	Tuesday and Thursday, 10:30–11:45 AM
Class Location	O'Neill 256
First Class	September 1, 2026
Last Class	December 10, 2026
Schedule Note	No class on October 13 (Monday class schedule followed) or November 26 (Thanksgiving)
Final Project Due	December 15, 2026

Texts and References

1. McNelis, Paul D. (2005), *Neural Networks in Finance: Gaining Predictive Edge in the Market*. Elsevier Academic Press.

Throughout the course, we refer to this text as **NNFin**.

2. James, Gareth, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor (2023), *An Introduction to Statistical Learning with Applications in Python*. Springer.

Throughout the course, we refer to this text as **ISLP**.

Several copies of these books will be available on reserve in the library. Additional journal articles, Python examples, and Jupyter notebooks will be distributed throughout the semester.

Course Description

This course introduces undergraduate students to modern machine-learning methods and their applications in economics and finance. Topics include forecasting, classification, dimensionality reduction, financial connectedness, and sequential decision making. Particular emphasis is placed on understanding how machine learning extends classical econometric methods and on evaluating predictive performance using out-of-sample evidence.

Students will work with real-world economic and financial data, including equity prices, exchange rates, interest rates, volatility measures, bank failures, loan defaults, customer churn, and portfolio-selection problems. The course begins with traditional forecasting methods and progressively introduces modern machine-learning techniques, including regularization, neural networks, tree-based methods, dimensionality reduction, network analysis, and reinforcement learning.

A central theme of the course is that predictive success should be judged primarily by out-of-sample performance rather than by in-sample fit.

Machine learning is a rapidly expanding field, and there is always more material available than can be covered in a single semester. My preference is that students learn a smaller number of methods thoroughly rather than encounter many techniques superficially. Accordingly, we will proceed at a pace that emphasizes understanding, implementation, and interpretation rather than breadth for its own sake.

Learning Objectives

By the end of the course, students should be able to:

1. Construct financial and economic forecasting datasets.
2. Estimate and evaluate classical forecasting models.
3. Understand volatility and risk-forecasting methods.
4. Apply machine-learning methods to regression and classification problems.
5. Use regularization methods such as LASSO for variable selection.
6. Implement cross-validation procedures for model selection.

7. Apply principal-component and nonlinear dimensionality-reduction methods.
8. Analyze financial interconnectedness using network methods.
9. Understand reinforcement learning as a framework for sequential decision making.
10. Evaluate competing models using out-of-sample performance measures.

Software

All empirical work will be conducted using Python and Jupyter notebooks.

Installing Python and Jupyter Notebook

Students should install Python and Jupyter Notebook on their laptop computers before the first class. For this course, **Python 3.13 (64-bit)** is recommended because it is compatible with the principal scientific-computing and machine-learning packages used in the course, including current TensorFlow releases.

1. Download and install Python.

Go to the official Python website: <https://www.python.org/downloads/>

Download a Python 3.13 installer for your computer.

- **Windows:** Download the 64-bit Windows installer. During installation, check the option **Add python.exe to PATH** if it is offered.
- **Mac:** Download the macOS installer from the Python website and complete the standard installation.

2. Check that Python is installed.

On Windows, open *Command Prompt* or *Windows Terminal* and enter:

```
py -version
```

On a Mac, open *Terminal* and enter:

```
python3 -version
```

The response should show Python 3.13.x.

3. Install Jupyter Notebook and the basic course libraries.

On Windows, enter:

```
py -m pip install -upgrade pip
py -m pip install notebook numpy pandas matplotlib scikit-learn statsmodels
xgboost
```

On a Mac, enter:

```
python3 -m pip install -upgrade pip
python3 -m pip install notebook numpy pandas matplotlib scikit-learn
statsmodels xgboost
```

TensorFlow/Keras will be installed later when it is needed in the course; separate instructions will be provided if necessary.

4. Start Jupyter Notebook.

In Command Prompt, Windows Terminal, or Mac Terminal, enter:

```
jupyter notebook
```

A web browser should open and display the Jupyter file browser. Students can then open a course notebook or create a new Python notebook.

5. Test the installation.

Create a new Python notebook and enter the following in the first cell:

```
print("Python and Jupyter are working.")
```

Run the cell. If the message appears beneath the cell without an error, the basic installation is working.

Official Jupyter installation information is available at <https://jupyter.org/install>.

The primary software libraries will include:

- NumPy
- Pandas
- Matplotlib
- Scikit-Learn
- Statsmodels
- TensorFlow/Keras
- XGBoost

Students should bring their laptop computers to class. Much of the learning will come from replicating, modifying, and extending examples discussed during lectures.

Students are welcome to use R, Julia, or Stata as alternatives. The responsible use of AI tools such as ChatGPT and Claude is encouraged, particularly for coding assistance and software troubleshooting.

Data Sources

Throughout the semester, we will work with publicly available economic and financial datasets. Primary sources include:

- Federal Reserve Economic Data (FRED)
- Yahoo Finance
- Stooq
- DBnomics
- Kaggle datasets
- U.S. Treasury and Bureau of Labor Statistics data

Course Outline

The course is organized into four parts covering five broad themes: forecasting, supervised machine learning, dimensionality reduction, financial networks, and sequential decision making.

Part I. Financial Data and Forecasting

1. Financial Data and Forecasting Fundamentals

- Asset prices, log returns, and volatility measures
- Realized volatility
- Data visualization
- Training and testing samples
- Recursive forecasting
- Rolling forecasting windows
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)

2. Classical Forecasting Methods

- OLS regression
- Lagged-return models
- Lagged-volatility models
- Rolling regressions
- Recursive estimation
- Kalman filtering

3. Volatility Forecasting of Asset-Market Returns

- ARCH and GARCH models
- Forecasting conditional volatility
- Realized volatility
- Comparing GARCH volatility with realized volatility
- Volatility persistence
- Leverage effects
- Forecast-accuracy assessment

Part II. Supervised Machine Learning

4. Function Approximation and Neural Networks

- Polynomial regression
- Chebyshev, Legendre, Hermite, and Laguerre polynomials
- Basis-function approximation
- Perceptrons
- Feedforward neural networks
- Backpropagation
- Hyperparameter selection
- Recurrent neural networks
- Long Short-Term Memory (LSTM) networks
- Sequence forecasting
- Return and volatility prediction

5. Classification and Supervised Learning

- Binary and multiclass classification
- Training and testing datasets
- Class imbalance
- Confusion matrices
- **Applications**
 - Titanic survival prediction
 - Texas bank-failure prediction
 - Brazilian automobile-loan default prediction
 - Customer-churn prediction
- **Classical classification methods**
 - Linear Probability Model
 - Logistic Regression, or Logit

- Probit models
- Odds ratios and marginal effects
- **Nearest-neighbor methods**
 - k -nearest neighbors, or k -NN
 - Distance metrics
 - Feature scaling
 - Choice of k
 - Curse of dimensionality
- **Neural-network classification**
 - Logistic activation function, or logsig
 - Binary cross-entropy loss
 - Feedforward neural networks
 - Classification boundaries
- **Tree-based methods**
 - Decision trees
 - Entropy and information gain
 - Gini impurity
 - Tree pruning
- **Ensemble and boosting methods**
 - Bagging
 - Random forests
 - Feature importance
 - Gradient boosting
 - XGBoost
- **Classification evaluation**
 - Accuracy
 - Precision
 - Recall
 - F1 score
 - ROC curves
 - Area Under the Curve (AUC)

6. Regularization and Model Selection

- Overfitting
- Bias–variance tradeoff
- Feature selection
- Ridge regression

- LASSO
- Elastic Net
- Sparse forecasting models
- **Cross-validation**
 - K-fold cross-validation
 - Time-series cross-validation
 - Rolling-window validation
 - Hyperparameter tuning
 - Selection of optimal penalty parameters

Part III. Dimensionality Reduction and Financial Networks

7. Dimensionality Reduction

- Curse of dimensionality
- Feature extraction
- Feature selection
- Visualization of high-dimensional data
- **Principal Components Analysis (PCA)**
 - Covariance matrices
 - Eigenvalues and eigenvectors
 - Variance explained
 - Economic and financial applications
- **Factor models**
 - Statistical factors
 - Macroeconomic factors
 - Financial factor models
- **Nonlinear dimensionality reduction**
 - Kernel PCA
 - Isomap
 - Locally Linear Embedding (LLE)
 - t-SNE
 - UMAP
 - Autoencoding

8. Financial Networks and Connectedness

- Vector Autoregressions (VAR models)
- Forecast Error Variance Decomposition
- Dynamic systems

- **Diebold–Yilmaz connectedness measures**

- Volatility spillovers
- Equity-market connectedness
- Exchange-rate connectedness
- Financial contagion
- Network visualization

- **Sparse network estimation**

- LASSO-VAR models
- High-dimensional systems
- Dynamic network analysis
- Systemic-risk measurement

Part IV. Reinforcement Learning and Portfolio Decision Making

- **Sequential decision making in portfolio choice**

- States
- Actions
- Rewards
- Episodes

- **Portfolio environment**

- Stock prices as state variables
- Portfolio weights as part of the state
- Discrete asset-allocation actions
- Reward as portfolio return

- **Policy-gradient reinforcement learning**

- Policy functions
- Neural-network policies
- Proximal Policy Optimization (PPO)
- Exploration versus exploitation

- **Financial applications**

- Dynamic portfolio allocation
- Buy-Hold-Sell decisions
- Stock selection
- Trading-strategy evaluation

- **Performance evaluation**

- Portfolio value over time
- Daily portfolio returns
- Cumulative returns
- Sharpe ratios
- Drawdowns
- Transaction costs

Model Evaluation

Throughout the course, competing forecasting and classification models will be evaluated using both in-sample and out-of-sample performance measures.

Forecasting Performance Measures

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Directional accuracy
- Out-of-sample R^2
- Diebold–Mariano forecast-comparison tests

Classification Performance Measures

- Confusion matrices
- Accuracy
- Precision
- Recall
- F1 score
- ROC curves
- Area Under the Curve (AUC)

Assignments: Empirical Projects

There will be three empirical papers, each approximately 5–10 pages, applying methods covered in class to similar or alternative datasets. Larger datasets are encouraged, both in the length of the time series and in cross-sectional breadth. Assignments may be co-authored with up to two other students, for groups of no more than three. Students are required to submit a précis one week before each paper is due.

Assignment	Due Dates	Weight
First Assignment	Précis: September 22; paper: September 29, 2026	20%
Second Assignment	Précis: November 3; paper: November 10, 2026	30%
Final Assignment	Précis: December 8; paper: December 15, 2026	30%

Illustrative topics include:

- Forecasting stock returns
- Forecasting market volatility
- Exchange-rate forecasting
- Neural-network forecasting of financial variables
- Titanic survival classification
- Texas bank-failure prediction
- Brazilian automobile-loan default prediction
- Customer-churn prediction
- Dimensionality reduction for financial and macroeconomic datasets
- Volatility connectedness among financial markets
- Sparse network estimation using LASSO-VAR models
- Reinforcement learning for stock selection using Buy-Hold-Sell decisions

Students are encouraged to propose their own project topics, subject to instructor approval.

Class Participation (20%)

Class participation includes the timely submission of progress reports on the three written assignments. The required précis for each assignment will serve as a progress report and should identify the data source or sources and the key question being investigated.

In addition, as part of the participation grade, each student is required, together with their group, to prepare a 5–10 minute video by November 1 for submission to Canvas. The video may be recorded jointly using Zoom and should summarize the key insights from the computational experiments, supported by clear and appropriate visualizations of the results. The videos will be reviewed by other members of the class. This is an opportunity for students to teach and learn from one another while also developing the ability to present and communicate computational work to a broader audience.

Course Policies

Attendance

- Regular attendance and participation are essential for success in this course.
- Students may occasionally need to miss class because of illness, family circumstances, athletic commitments, or religious observances. All classes will be recorded whenever feasible.
- Except in emergencies, students should notify their team leader or the instructor in advance when they are unable to attend class.
- Students should arrive before the beginning of class. Occasional late arrivals are understandable, but chronic tardiness is disruptive to the learning environment.

Classroom Conduct

- Coffee and soft drinks are permitted in class. Please avoid using class time as a meal period.
- Laptop computers are encouraged for course-related work. Activities unrelated to the course, including social media and general internet browsing, can be distracting to both the user and fellow students.
- Courtesy, professionalism, and respect for others are expected at all times.

Late Assignments

- Meeting deadlines is an important professional skill, and students are expected to submit assignments on time.
- Extensions may be granted for reasonable circumstances when requested in advance.
- Requests for extensions should be made in writing whenever possible.

Academic Integrity and Artificial Intelligence

- The responsible use of AI tools is permitted and encouraged.
- Boston College has a contractual agreement with Gemini (which you can access through your email account) and Copilot, for data privacy and protection.
- Boston College policies and guidance on the use of generative AI can be found at [Boston College Generative AI: AI Tools and Policies](#).
- Students remain responsible for verifying the accuracy of all code, results, references, and written material produced with AI assistance.
- Written reports and project submissions must reflect the student's own understanding and analysis.
- All sources, including AI-assisted work, should be properly documented when appropriate.

- Students should be aware that generative AI systems can produce incorrect or fabricated information.
- Boston College's Academic Integrity Policy applies fully to all work submitted in this course.

Disability Accommodations

Boston College is committed to providing equal access to educational opportunities for all students. Students seeking accommodations should contact the appropriate university office as early as possible in the semester so that reasonable arrangements can be made.

Welcome to ECON 3389.3! I look forward to an engaging semester of computational exploration in macroeconomics.